

**HostelHub**

# Security, Data Protection & Infrastructure Whitepaper

Comprehensive Guide to Encryption, Cloud Servers, Database Hosting & Web-to-Mobile App Architecture

## DOCUMENT SCOPE

Data Security, Encryption & Cloud Hosting Architecture

## API APPLICATION SERVER

Render Cloud Platform (HTTPS Node.js Instance)

## ENCRYPTION STANDARD

bcryptjs (Salt 10) + JWT RSA/HS256 Bearer Tokens

## DATABASE CLOUD HOST

Supabase Managed PostgreSQL (AWS Infra)

## MOBILE NATIVE CONVERSION

Capacitor v8 Android Bridge (WebView Runtime)

## AUTHOR ENGINEERING TEAM

Abhinav Kumar (Lead Full-Stack Developer)

# 1. Data Protection & Security Architecture

HostelHub enforces multi-layered defense-in-depth security policies to ensure that student, warden, and staff data remains 100% confidential, tamper-proof, and safe from unauthorized access.

 **Zero Plaintext Storage:** All user passwords are encrypted using **bcryptjs hashing with 10 salt rounds** before hitting the database. Even database administrators cannot view original plaintext passwords.

## 1.1 Security Mechanisms Overview

| SECURITY LAYER     | TECHNOLOGY USED                | HOW IT PROTECTS YOUR DATA                                                                           |
|--------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|
| Password Security  | bcryptjs (10 Salt Rounds)      | One-way cryptographic hashing protects credentials against database leak attacks.                   |
| API Authentication | JSON Web Tokens (JWT)          | Stateless bearer tokens verify every incoming HTTP API call with role payload verification.         |
| Data in Transit    | HTTPS / TLS 1.3 Encryption     | All communication between mobile app and server is encrypted using SSL/TLS encryption.              |
| Database Safety    | Sequelize ORM Parameterization | Prevents SQL Injection attacks by automatically parameterizing and escaping raw database queries.   |
| HTTP Headers Guard | Helmet.js Middleware           | Secures Express backend by setting XSS Filter, Strict-Transport-Security, and MIME sniffing guards. |
| Native App Storage | Capacitor Preferences API      | Tokens & session state stored in Android native SharedPreferences, protected from WebView clearing. |

# 2. Database Hosting & Cloud Infrastructure

HostelHub relies on enterprise-grade cloud hosting providers to deliver 99.9% uptime and reliable automated data persistence.

### 🌐 Cloud Server Breakdown:

- **Database Server:** Hosted on **Supabase Cloud Managed PostgreSQL**. Supabase runs on top of secure Amazon Web Services (AWS) data centers with automatic daily backups and encrypted disk storage.
- **Backend REST API Server:** Hosted on **Render.com Cloud Application Platform** (<https://hostelhub-0cyi.onrender.com>). Render handles HTTP/HTTPS routing, Node.js process management, and SSL certificate renewals.
- **Media & Attachment Storage:** Complaint issue photos, work completion proofs, and profile pictures are uploaded to cloud buckets with fallback local path mapping.

```
[ Mobile Android App / Browser ]
    | (Encrypted HTTPS / WebSockets)
    ▼
[ Render.com Cloud API Server ] ← Node.js / Express Middleware
    | (TLS Encrypted Connection)
    ▼
[ Supabase PostgreSQL Cloud Database ] ← AWS Secured Infrastructure
```

### 3. Web-to-Native Mobile App Conversion (Capacitor Architecture)

HostelHub was built as an Angular Progressive Single Page Application and converted into a **Native Android APK** using **Capacitor v8** (developed by the Ionic engineering team).

#### 3.1 How Web Code Runs in Native APK

1. **Production Build Compilation:** The Angular source code ('TypeScript', 'HTML', 'CSS') is compiled into optimized JavaScript bundles inside 'dist/frontend-user'.
2. **Native Asset Syncing:** Capacitor ('npx cap sync') copies the web build directly into the Android native project assets folder ('android/app/src/main/assets/public').
3. **WebView Execution:** Android's high-performance native WebView engine loads the app locally inside the native APK wrapper, rendering at 60 FPS without needing external browser navigation bars.
4. **Native Bridge Hardware APIs:**
  - `@capacitor/camera` : Direct access to phone camera hardware for taking complaint attachment photos.
  - `@capacitor/preferences` : Access to Android native SharedPreferences for persistent authentication tokens.
  - `@capacitor/app` : Hardware back-button event listener enforcing the 2-step exit modal guard.

### 4. Critical Business & Security Rules

 **Profile Re-Approval Security Guard:** Standard profile fields (Name, Phone, Bio, Room Number, Profile Picture) save instantly. However, if a student attempts to edit **Gender** or **Academic Batch**, the system triggers a critical warning popup and requires Warden Re-approval ( `reApprovalStatus = true` ) before the changes take effect.

 **Dual-Press Back Exit Guard:** Pressing the Android hardware back button inside the app navigates back to the Home tab first. Pressing back a second time opens a clean confirmation modal asking **"Are you sure you want to exit HostelHub?"** with Yes/No options.